Notes: Cohen, Malloy, and Nguyen (Journal of Finance, 2020)
Lazy Prices

Contents

1 Big picture

- **Question.** Do changes in firms' annual and quarterly filings contain value-relevant information, and do investors promptly incorporate that information into prices?
- **Main result.** Firms that substantially change filing language subsequently underperform firms that do not change, while filing-date announcement returns are close to zero.
- **Mechanism.** Investors are inattentive to simple but subtle changes across long, complex corporate reports.
- **Economic takeaway.** Public disclosure can be informative yet underprocessed; information availability is not the same as information incorporation.

2 Incremental contribution of this paper

- Relative to disclosure work, the paper shows that declining announcement effects do not mean reports are uninformative; instead, investors miss information at release.
- Relative to textual-analysis work, it studies year-over-year changes rather than static tone or readability.
- Relative to underreaction work, it documents drift without an initial filing-date reaction.
- Relative to attention work, it uses SEC EDGAR download logs to measure whether investors compare current and prior filings.
- Relative to information theory, it updates the Grossman-Stiglitz intuition for a world in which collecting information is cheap but processing information is costly.

3 Data and measurement

The sample contains 10-K and 10-Q filings from SEC EDGAR from 1995 to 2014. The paper removes tables, HTML/XBRL tags, exhibits, ASCII PDFs, graphics, Excel files, and other non-text elements. It then compares each filing to the relevant prior-year filing.

$$\text{Cosine}(D_1, D_2) = \frac{D_1 \cdot D_2}{\|D_1\|\|D_2\|}, \qquad \text{Jaccard}(D_1, D_2) = \frac{|D_1 \cap D_2|}{|D_1 \cup D_2|}.$$

$$\text{Simple} = \frac{c_{\max} - c}{c_{\max}}, \qquad c = \frac{\text{additions} + \text{deletions} + \text{changes}}{(\text{Size}(D_1) + \text{Size}(D_2))/2}.$$

Interpretation. Higher similarity means less change; lower similarity means more change.

4 Models and empirical specifications

4.1 Calendar-time portfolios

For each similarity measure, firms are sorted into quintiles using the prior year's distribution. Firms enter portfolios in the month after the filing release and are held for three months. Q1 contains changers; Q5 contains nonchangers. The main portfolio is Q5–Q1.

4.2 Fama-MacBeth predictability

$$R_{i,t+1} = a_t + b_t \text{Similarity}_{i,t} + \Gamma_t' X_{i,t} + \varepsilon_{i,t+1}.$$

4.3 Attention interaction

$$R_{i,t+1} = a + b\text{Similarity}_{i,t} + c\text{Attention}_{i,t} + d(\text{Similarity}_{i,t} \times \text{Attention}_{i,t}) + \Gamma' X_{i,t} + \varepsilon_{i,t+1}.$$

Interpretation. If inattention drives the anomaly, the interaction should weaken return predictability when more investors compare current and prior filings.

5 Identification and empirical design

The paper uses multiple text measures, portfolio tests, Fama-MacBeth regressions, event-time returns, section-level analysis, EDGAR attention proxies, comparative-phrase tests, and real-outcome regressions. The triangulation points to delayed investor processing of textual disclosure changes.

6 Tables and figures

6.1 Figure 1: Length of 10-Ks and changes to 10-Ks over time; Table I: Summary statistics

Read: 10-Ks grow dramatically longer, and the number of textual changes also rises. Table I shows large filing samples and correlations among similarity measures.

Economic message. Disclosure becomes more complex, increasing comparison costs.

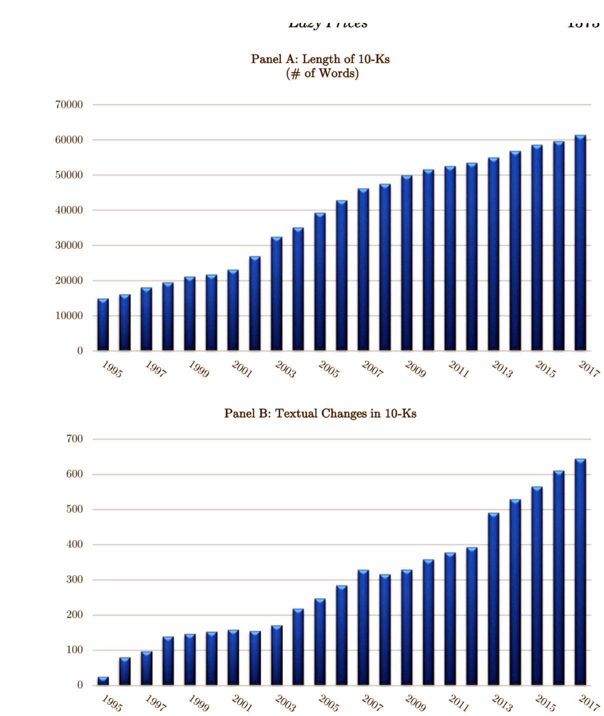


Figure 1. Length of 10-Ks and changes to 10-Ks over time. (Color figure can be viewed at wiley-

(a) Figure 1: length and changes

Table I

Summary Statistics for Firms' 10-Ks and 10-Qs

Panel A reports summary statistics for 10-Ks and 10-Qs over the period 1995 to 2014. *Document Size* is the number of words in each document. *Sentiment of Change* is the number of negative words in *Change* normalized by the number of words in *Change*, where *Change* is additions, deletions, and other changes identified from the function "diff" in Unix/Linux or the function "Track Changes" in Microsoft Word. *Uncertainty of Change* and *Litigiousness of Change* are the number of words categorized as uncertainty and litigiousness, respectively, normalized by the size of *Change*. *Change CEO* and *Change CFO* are indicator variables equal to 1 if the 10-K or 10-Q mentions a change in CEO or CFO, respectively. Sentiment category identifiers (e.g., negative, uncertainty, litigious) come from Loughran and McDonald's (2011) Master Dictionary. Panel B reports the summary statistics of four different measures of document similarity. Panel C reports the correlations between the four similarity measures used in this paper. *Sim_Cosine* is the cosine similarity measure, *Sim_Jaccard* is the Jaccard similarity measure, *Sim_MinEdit* is the minimum edit distance similarity measure, and *Sim_Simple* is the simple side-by-side comparison. Details on how we compute the four similarity measures can be found in the Internet Appendix

Panel A: Summary Statistics of Document Charateristics

	Count	Mean	*SD*	1%	50%	99%
Document Size—10-K	86,965	44,508.81	36,479	7,573	35,787	180,388
Document Size—10-Q	258,271	15,805.9	20,542.78	1,327	10,674	97,521
Sentiment of Change	345,639	0.07736	0.0179074	0	0.000146	0.003503
Uncertainty of Change	345,639	0.0005234	0.0110212	0	0.0001286	0.0026464
Litigiousness of Change	345,639	0.0009594	0.016019	0	0.0000668	0.0051982
Change CEO	345,639	0.0556158	0.2291785	0	0	1
Change CFO	345,639	0.0242542	0.1538377	0	0	1

Panel B: Summary Statistics of Similarity Measures

	Count	Mean	*SD*	1%	50%	99%
Sim_Cosine	327,130	0.8721032	0.1910398	0.1367042	0.947125	0.9951641
Sim_Jaccard	327,130	0.3948525	0.190596	0.0364943	0.4108108	0.765858
Sim_MinEdit	327,130	0.3763384	0.1714118	0.0516403	0.3927964	0.7649283
Sim_Simple	327,130	0.1464663	0.0927251	0.0427717	0.1171773	0.4283921

Panel C: Correlation

	Sim_Cosine	*Sim_Jaccard*	*Sim_MinEdit*	*Sim_Simple*
Sim_Cosine	1.0000			
Sim_Jaccard	0.6049	1.0000		
Sim_MinEdit	0.5031	0.7921	1.0000	
Sim_Simple	0.2076	0.4815	0.5834	1.0000

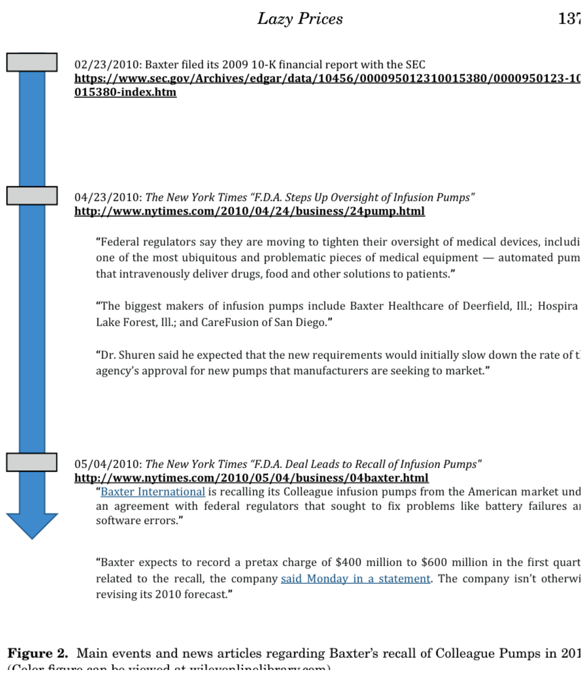
"a number," lines with only "number" + "title names," etc., while also ensuring that each schedule is captured exactly once. We repeat this process many

(b) Table I: summary statistics

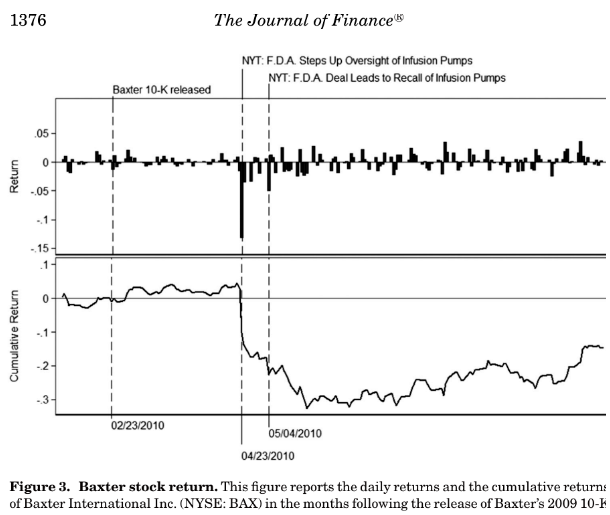
6.2 Figures 2–5: Baxter example

Read: Baxter's filing language changes predate the later FDA-related news and stock-price decline. Keyword counts and passage highlights show how subtle changes map into economic risk.

Economic message. The example illustrates delayed price discovery from filing changes.



(a) Figure 2: Baxter news events



(b) Figure 3: Baxter stock return

ational Inc. (NYSE: BAX) in the months following the release of Baxt

Word counts	2007 10-K	2008 10-K	2009 10-K
FDA	33	28	48
Recall	16	20	30
Colleague Pump	29	28	79

(a) Figure 4: important keywords

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Example 1	
2008:	2009:
With respect to COLLEAGUE, the company remains in active dialogue with the FDA about various matters, including the company's remediation plan and reviews of the Company's facilities, processes and quality controls by the company's outside expert pursuant to the requirements of the company's Consent Decree. The outcome of these discussions with the FDA is uncertain and may impact the nature and timing of the company's actions and decisions with respect to the COLLEAGUE pump. The company's estimates of the costs related to these matters are based on the current remediation plan and information currently available. It is possible that additional charges related to COLLEAGUE may be required in future periods, based on new information, changes in estimates, and modifications to the current remediation plan as a result of ongoing dialogue with the FDA.	The company remains in active dialogue with the FDA regarding various matters with respect to the company's COLLEAGUE infusion pumps, including the company's remediation plan and reviews of the company's facilities, processes and quality controls by the company's outside expert pursuant to the requirements of the company's Consent Decree. The outcome of these discussions with the FDA is uncertain and may impact the nature and timing of the company's actions and decisions with respect to the COLLEAGUE pump. The company's estimates of the costs related to these matters are based on the current remediation plan and information currently available. It is possible that substantial additional charges, including significant asset impairments, related to COLLEAGUE may be required in future periods, based on new information, changes in estimates, and modifications to the current remediation plan.

Example 2	
2008:	2009:
In the third quarter of 2008, as a result of the company's decision to upgrade the global pump base to a standard software platform and other changes in the estimated costs to execute the remediation plan, the company recorded a charge of $72 million. This charge consisted of $46 million for cash costs and $26 million principally relating to asset impairments and inventory used in the remediation plan. The reserve for cash costs primarily consisted of costs associated with the deployment of the new software and additional repair and warranty costs. The following summarizes cash activity in the company's COLLEAGUE and SYNDEO infusion pump reserves through December 31, 2008.	In the third quarter of 2008, as a result of the company's decision to upgrade the global pump base to a standard software platform and other changes in the estimated costs to execute the remediation plan, the company recorded a charge of $72 million. This charge consisted of $46 million for cash costs and $26 million principally relating to asset impairments and inventory used in the remediation plan. The reserve for cash costs primarily consisted of costs associated with the deployment of the new software and additional repair and warranty costs. In 2009, the company recorded a charge of $27 million related to planned retirement costs associated with SYNDEO and additional costs related to the COLLEAGUE infusion pump. This charge consisted of $14 million for cash costs and $13 million related to asset impairments. The reserve for cash costs primarily related to customer accommodations and additional warranty costs. The charges were recorded in cost of sales in the company's consolidated statements of income, and were included in the Medication Delivery segment's pre-tax income. The following summarizes cash activity in the company's COLLEAGUE and SYNDEO infusion pump reserves through December 31, 2009.

Figure 5. Example passages and the changes made to them from Baxter's 10-Ks in 2008 and 2009. (Color figure can be viewed at wileyonlinelibrary.com)

Circling back, would being attentive to the changes in Baxter's 10-K have made a difference to investors in the company? Going back to Figure 3, the answer appears to be yes. Not only did the price of Baxter not move at all around the public filing of the 2009 10-K (February 23, 2010), but it did not move for the next two months—until the news report by the *New York Times* on April 23. Reading and reacting to these negative changes by shorting Baxter at any point in the two months leading up to the *New York Times*

(b) Figure 5: passage changes

6.3 Figure 6 and Table II: Section definitions and main calendar-time returns

Read: Figure 6 defines sections. Table II shows that nonchangers outperform changers across similarity measures.

Economic message. The main asset-pricing result is that document change predicts future underperformance.

Form 10-K	
Item 1	Business
Item 1A	Risk Factors
Item 2	Properties
Item 3	Legal Proceedings
Item 4	Mine Safety Disclosures
Item 5	Market for Registrant's Common Equity, Related Stockholder Matters and Issuer Purchases of Equity Securities
Item 6	Selected Financial Data
Item 7	Management's Discussion and Analysis of Financial Condition and Results of Operations
Item 7A	Quantitative and Qualitative Disclosures About Market Risk
Item 8	Financial Statements and Supplementary Data
Item 9	Changes in and Disagreements With Accountants on Accounting and Financial Disclosure
Item 9A	Controls and Procedures
Item 9B	Other Information
Item 10	Directors, Executive Officers and Corporate Governance
Item 11	Executive Compensation
Item 12	Security Ownership of Certain Beneficial Owners and Management and Related Stockholder Matters
Item 13	Certain Relationships and Related Transactions, and Director Independence
Item 14	Principal Accounting Fees and Services

Form 10-Q	
Item 1	Financial Statements
Item 2	Management's Discussion and Analysis of Financial Condition and Results of Operations
Item 3	Quantitative and Qualitative Disclosures About Market Risk
Item 4	Controls and Procedures
Item 21	Legal Proceedings
Item 21A	Risk Factors
Item 22	Unregistered Sales of Equity Securities and Use of Proceeds
Item 23	Defaults Upon Senior Securities
Item 24	Mine Safety Disclosures
Item 25	Other Information

Figure 6. Section definitions in 10-Ks and 10-Qs.

(a) Figure 6: section definitions

Table II
Main Results—Calendar-Time Portfolio Returns

This table reports calendar-time portfolio returns. Portfolio returns are multiplied by 100. Sim_*Cosine* is the cosine similarity measure, *Sim_Jaccard* is the Jaccard similarity measure, *Sim_MinEdit* is the minimum edit distance similarity measure, and *Sim_Simple* is the simple side-by-side comparison. For each of the four similarity measures, we compute quintiles based on the prior year's distribution of similarity measures across all stocks. Stocks then enter the quintile portfolios in the month after the public release of one of their 10-K or 10-Q reports. Stocks are held in the portfolio for three months. We report excess returns (return minus risk-free rate), Fama-French three-factor alphas (market, size, and value), and five-factor alphas (market, size, value, momentum, and liquidity). Panel A reports equal-weighted portfolio returns, and Panel B reports value-weighted portfolio returns. *t*-Statistics are reported below the estimates. Statistical significance at the 1%, 5%, and 10% levels is indicated by ***, **, and *, respectively

Panel A: Equally Weighted

	Sim_Cosine						*Sim_Jaccard*					
	Q1	Q2	Q3	Q4	Q5	Q5 – Q1	Q1	Q2	Q3	Q4	Q5	Q5 – Q1
Excess return	0.63*	0.72*	0.72**	0.85**	0.92***	0.31***	0.59	0.67*	0.69*	0.82**	0.98***	0.38***
	(1.68)	(1.96)	(2.11)	(2.59)	(2.80)	(3.13)	(1.48)	(1.74)	(1.89)	(2.35)	(3.01)	(2.65)
Three-factor alpha	−0.15**	−0.08	−0.05	0.09	0.18***	0.34***	−0.16**	−0.10	−0.06	0.08	0.28***	0.44***
	(−2.19)	(−1.10)	(−0.72)	(1.21)	(2.66)	(4.45)	(−1.99)	(−1.22)	(−0.81)	(1.05)	(3.47)	(4.56)
Five-factor alpha	−0.12*	−0.05	−0.04	0.10	0.21***	0.32***	−0.14*	−0.07	−0.06	0.09	0.28***	0.42***
	(−1.75)	(−0.74)	(−0.53)	(1.29)	(3.28)	(4.21)	(−1.84)	(−0.93)	(−0.86)	(1.19)	(3.57)	(4.31)
	Sim_MinEdit						*Sim_Simple*					
	Q1	Q2	Q3	Q4	Q5	Q5 – Q1	Q1	Q2	Q3	Q4	Q5	Q5 – Q1
Excess return	0.61	0.66*	0.70*	0.86**	0.99***	0.36***	0.72*	0.79**	0.82**	0.90***	0.90***	0.18
	(1.60)	(1.78)	(1.94)	(2.58)	(3.36)	(2.69)	(1.87)	(2.12)	(2.34)	(2.73)	(3.04)	(1.20)
Three-factor alpha	−0.19**	−0.14*	−0.10	0.10	0.30***	0.48***	−0.08	−0.02	0.03	0.14**	0.20**	0.28***
	(−2.56)	(−1.91)	(−1.52)	(1.37)	(4.00)	(5.96)	(−1.09)	(−0.21)	(0.38)	(2.01)	(2.57)	(3.22)
Five-factor alpha	−0.15**	−0.11	−0.08	0.12*	0.30***	0.45***	−0.06	0.03	0.04	0.16**	0.21***	0.27***
	(−2.14)	(−1.59)	(−1.31)	(1.70)	(4.11)	(5.46)	(−0.89)	(0.37)	(0.63)	(2.30)	(2.68)	(3.01)

(Continued)

(b) Table II, Panel A

Table II—*Continued*

Panel B: Value Weighted

	Sim_Cosine						*Sim_Jaccard*					
	Q1	Q2	Q3	Q4	Q5	Q5 – Q1	Q1	Q2	Q3	Q4	Q5	Q5 – Q1
Excess return	0.43	0.47	0.55*	0.73**	0.78**	0.34**	0.23	0.32	0.48	0.61*	0.79**	0.56***
	(1.32)	(1.45)	(1.74)	(2.35)	(2.40)	(2.53)	(0.64)	(0.88)	(1.33)	(1.84)	(2.47)	(3.75)
Three-factor alpha	−0.15*	−0.15*	−0.04	0.10	0.20*	0.35***	−0.32***	−0.21	−0.09	0.07	0.23**	0.54***
	(−1.84)	(−1.79)	(−0.49)	(1.17)	(1.97)	(2.63)	(−2.97)	(−1.30)	(−0.73)	(0.60)	(2.01)	(4.08)
Five-factor alpha	−0.12	−0.19**	−0.06	0.12	0.23**	0.34**	−0.23**	−0.17	−0.07	0.13	0.23**	0.46***
	(−1.38)	(−2.13)	(−0.64)	(1.36)	(2.23)	(2.53)	(−2.20)	(−1.04)	(−0.59)	(1.18)	(2.11)	(3.44)
	Sim_MinEdit						*Sim_Simple*					
	Q1	Q2	Q3	Q4	Q5	Q5 – Q1	Q1	Q2	Q3	Q4	Q5	Q5 – Q1
Excess return	0.42	0.45	0.62*	0.76**	0.83***	0.39**	0.24	0.61*	0.77**	0.78**	0.74**	0.50***
	(1.25)	(1.38)	(1.88)	(2.42)	(2.92)	(2.31)	(0.69)	(1.88)	(2.45)	(2.53)	(2.48)	(2.69)
Three-factor alpha	−0.18**	−0.16*	−0.01	0.17*	0.28**	0.46***	−0.39***	0.02	0.18*	0.19*	0.19	0.58***
	(−2.29)	(−1.91)	(−0.14)	(1.74)	(2.49)	(3.06)	(−3.89)	(0.18)	(1.87)	(1.88)	(1.45)	(3.59)
Five-factor alpha	−0.17**	−0.14*	0.00	0.17*	0.21*	0.37**	−0.36***	0.05	0.18*	0.18*	0.15	0.51***
	(−2.02)	(−1.67)	(0.04)	(1.78)	(1.84)	(2.45)	(−3.49)	(0.66)	(1.78)	(1.71)	(1.15)	(3.14)

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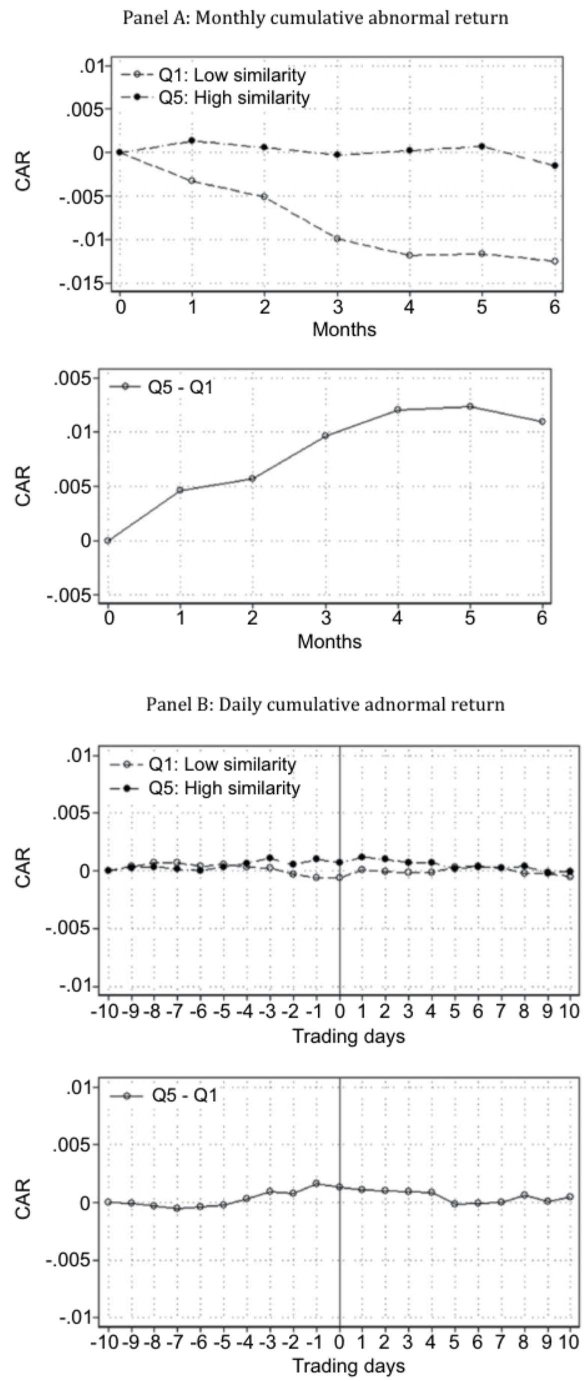
(a) Table II, Panel B



e **returns.** This figure plots weighted-average cumu
est similarity) and bottom (lowest similarity) quintile
ı the prior year's distribution of Jaccard similarity a
urns adjusted for market returns. Event dates are d
-K or 10-Q. Panel A shows the weighted-average

(b) Figure 7: event-time returns

6.4 Tables III–VI: Trading characteristics, Fama-MacBeth tests, mechanisms, and sentiment controls

Read: Table III evaluates trading characteristics and limits to arbitrage. Table IV shows continuous return predictability. Table V links changes to negative/uncertain/legal content. Table VI shows similarity remains predictive after sentiment controls.

Economic message. The anomaly is not purely a tone or illiquidity artifact.

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Table III
Characteristics of Quintile Portfolios

This table reports mean *Market Value of Equity* (determined at the end of the months in which the quintile cut-off points are determined, in $1,000 dollars), *Monthly Turnover*, *Shorting Fees*, and *Sentiment of Changes* of the five quintile portfolios.

	Q1	Q2	Q3	Q4	Q5
Market Value of Equity	3,507,587	3,219,430	2,829,955	2,504,717	2,464,603
Monthly Turnover	0.0663	0.0850	0.0804	0.0867	0.0706
Shorting Fees (bps)	71.6958	80.6361	92.0500	87.0690	73.5453
Sentiment of Changes	0.0016	0.0008	0.0006	0.0005	0.0004

composition and characteristics of both sides of the L/S portfolio. For example, it could be the case that the short side simply contains a set of smaller firms that are difficult (and expensive) to short. Or perhaps there is not a significant turnover of small or illiquid stocks to trade. Both of these possibilities might make the returns we document fall within simple limits of arbitrage. Table III presents the average size, turnover, shorting costs (in basis points), and sentiment (as defined in Table I) for all five quintile portfolios. As can be seen, there

(a) Table III: characteristics

Table IV
Main Results—Fama-MacBeth Regressions

This table reports results of Fama-MacBeth cross-sectional regressions of individual firm-level stock returns on our four similarity measures and a number of known return predictors. Return, the dependent variable, is multiplied by 100. *Sim_Cosine* is the cosine similarity measure, *Sim_Jaccard* is the Jaccard similarity measure, *Sim_MinEdit* is the minimum edit distance similarity measure, and *Sim_Simple* is the simple side-by-side comparison. *Size* is log of market value of equity, *log(BM)* is log book value of equity over market value of equity, *Ret*(–1,0) is the previous month's return, and *Ret*(–12,–1) is the cumulative return from month –12 to month –1. *SUE* is the standardized unexpected earnings and computed as actual earnings per share minus average analyst forecast earnings per share, divided by the standard deviation of forecasts. *t*-Statistics are reported below the estimates. Statistical significance at the 1%, 5%, and 10% levels is indicated by ***, **, and *, respectively

	Ret											
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
Sim_Cosine	0.45***	0.31**	0.37**									
	(2.65)	(2.51)	(2.18)									
Sim_Jaccard				0.82***	0.66***	0.59***						
				(3.26)	(3.82)	(3.41)						
Sim_MinEdit							0.54**	0.41***	0.29**			
							(2.54)	(2.78)	(2.00)			
Sim_Simple										0.04**	0.03**	0.03**
										(2.10)	(2.25)	(2.11)
Size		0.00	0.00		0.01	0.01		0.01	0.01		0.01	0.00
		(0.11)	(0.05)		(0.25)	(0.11)		(0.26)	(0.10)		(0.24)	(0.05)
log(BM)		0.17*	0.16*		0.17*	0.16*		0.17*	0.16*		0.17*	0.16*
		(1.89)	(1.71)		(1.88)	(1.70)		(1.90)	(1.72)		(1.87)	(1.70)
Ret(–1,0)		−0.03***	−0.02***		−0.03***	−0.02***		−0.03***	−0.02***		−0.03***	−0.02***
		(−3.93)	(−3.68)		(−3.97)	(−3.70)		(−3.97)	(−3.69)		(−3.99)	(−3.71)
Ret(−12,−1)		0.64**	0.36		0.64**	0.36		0.64**	0.36		0.64**	0.37
		(2.34)	(1.25)		(2.34)	(1.25)		(2.34)	(1.24)		(2.35)	(1.29)
SUE			0.07***			0.07***			0.07***			0.07***
			(6.56)			(6.54)			(6.56)			(6.60)
Cons	0.58	0.58	0.67	0.64	0.46	0.69	0.76**	0.57	0.84	−0.02	−0.02	−0.01
	(1.45)	(0.67)	(0.57)	(1.64)	(0.52)	(0.58)	(1.98)	(0.64)	(0.71)	(−1.31)	(−1.02)	(−0.71)
R^2	0.00	0.04	0.05	0.00	0.04	0.05	0.00	0.04	0.05	0.00	0.04	0.05
N	713,451	713,451	496,084	713,451	713,451	496,084	713,451	713,451	496,084	713,680	713,680	495,931

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(b) Table IV: Fama-MacBeth

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Table V
Potential Mechanism

This table explores the potential mechanism behind our results. We regress our similarity measure on a host of characteristics describing the document in question. *Sim_Simple*, the dependent variable, is multiplied by 100. *Sentiment of Change* is the number of negative words in *Change* normalized by the number of words in *Change*, where *Change* is computed as the number of additions, deletions, and other changes identified from the function "diff" in Unix/Linux or the function "Track Changes" in Microsoft Word. *Uncertainty of Change* and *Litigiousness of Change* are the number of words categorized as uncertainty and litigiousness, respectively, normalized by the size of *Change*. *Change CEO* and *Change CFO* are indicator variables equal to 1 if the 10-K or 10-Q mentions a change in CEO or CFO, respectively. Sentiment category identifiers (e.g., negative, uncertainty, and litigious) come from Loughran and McDonald's (2011) Master Dictionary. All regressions include firm and month fixed effects. Standard errors are clustered at the firm level. *t*-Statistics are reported below the estimates. Statistical significance at the 1%, 5%, and 10% levels is indicated by ***, **, and *, respectively.

	Sim_Simple				
	(1)	(2)	(3)	(4)	(5)
Sentiment of Change	−2.49***				
	(−37.83)				
Uncertainty of Change		−3.57***			
		(−34.15)			
Litigiousness of Change			−0.12**		
			(−2.11)		
Change CEO				−0.01***	
				(−7.10)	
Change CFO					−0.01***
					(−5.75)
Cons	0.18***	0.19***	0.18***	0.18***	0.18***
	(28.52)	(17.40)	(17.25)	(17.31)	(17.29)

(a) Table V: potential mechanism

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Table VI
Fama-MacBeth Regressions, Controlling for Sentiment and Document Size

This table reports results of Fama-MacBeth cross-sectional regressions of individual firm-level stock returns on our *Sim_Jaccard* similarity measures and a number of known return predictors. *Return*, the dependent variable, is multiplied by 100. *Sim_Jaccard* is the Jaccard similarity measure. *Sentiment of Change is Positive* is the number of positive words in *Change*, normalized by the size of *Change*, where *Change* is additions, deletions, and other changes identified from the function "diff" in Unix/Linux or the function "Track Changes" in Microsoft Word. *Size* is log of market value of equity, *log(BM)* is log book value of equity over market value of equity, *Ret*(–1,0) is previous month's return, and *Ret*(–12,–1) is the cumulative return from month –12 to month –1. *Log(Document Size)* is the logarithm of the number of words in a document. Δ *Log(Document Size)* is the quarter-on-quarter change in *Log(Document Size)*. *t*-Statistics are reported below the estimates. Statistical significance at the 1%, 5%, and 10% levels is indicated by ***, **, and *, respectively.

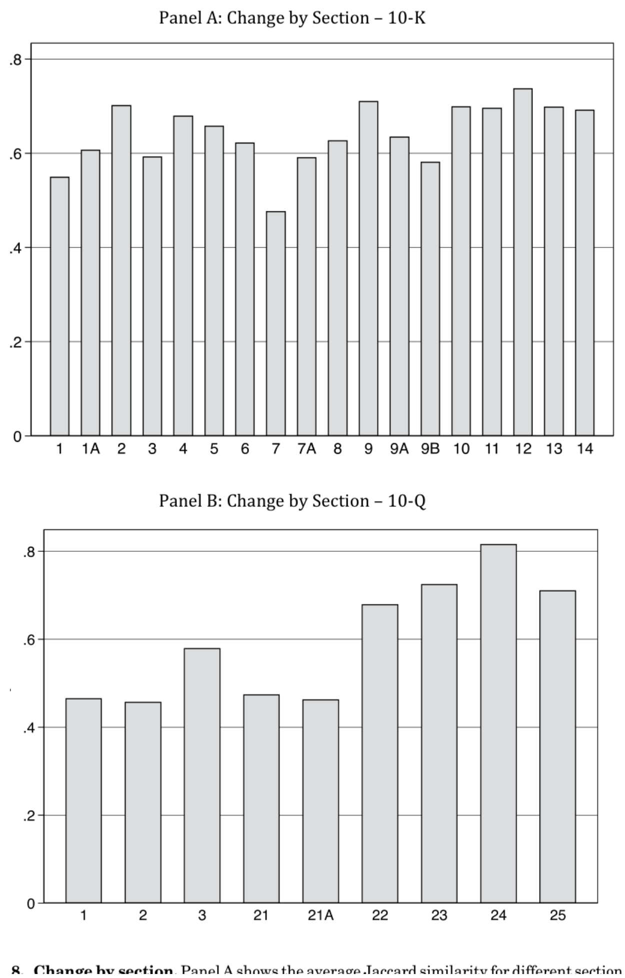
	Ret		
	(1)	(2)	(3)
Sim_Jaccard	0.57***	0.58***	0.58***
	(3.45)	(3.78)	(3.82)
Sentiment of Change is Positive	0.19***	0.21***	0.21***
	(3.85)	(4.21)	(4.33)
Log(Document Size)		0.01	0.03
		(0.65)	(1.40)
Δ *Log(Document Size)*			−0.41**
			(−2.30)
Size	0.00	0.00	−0.00
	(0.10)	(0.07)	(−0.01)
log(BM)	0.17	0.16	0.16
	(1.64)	(1.5858)	(1.5471)
Ret(−1,0)	−0.03***	−0.03***	−0.03***
	(−4.15)	(−4.19)	(−4.20)
Ret(−12, −1)	0.74***	0.74***	0.74***
	(2.71)	(2.70)	(2.69)
Cons	0.55	0.41	0.25

(b) Table VI: sentiment controls

6.5 Figure 8, Table VII, and Figure 9: Section-level changes and alphas

Read: MD&A changes most often, but Risk Factors and other economically sensitive sections produce especially strong return predictability.

Economic message. Investors especially miss changes in risk- and litigation-related narrative sections.



(a) Figure 8: change by section

Table VII

Portfolio Sorts—By Document Section

This table reports calendar-time portfolio returns for the common sections of firms' 10-K and 10-Q financial reports: Management's Discussion and Analysis, Legal Proceedings, Quantitative and Qualitative Disclosures about Market Risk, Risk Factors, and Other Information. Portfolio returns are multiplied by 100. Similarity measures for each section are computed using only the textual portion in that section. For each of the four similarity measures, we compute quintiles based on the prior year's distribution of similarity measures across all stocks. Stocks then enter the quintile portfolio in the month after the public release of one of their 10-K or 10-Q reports. Firms are held in the portfolio for three months. We report Excess Returns (return minus risk free rate), Fama-French three-factor alphas (market, size, and value), and five-factor alphas (market, size, value, momentum, and liquidity) of the top minus bottom quintile portfolio (Q5 – Q1). Panel A reports equal-weighted portfolio returns, and Panel B reports value-weighted portfolio returns. *t*-Statistics are reported below the estimates. Statistical significance at the 1%, 5%, and 10% levels is indicated by ***, **, and *, respectively.

Panel A: Equally Weighted

	Sim_Cosine			*Sim_Jaccard*		
	Excess Return	Three-Factor Alpha	Five-Factor Alpha	Excess Return	Three-Factor Alpha	Five-Factor Alpha
Management's Discussion and Analysis	0.13	0.11*	0.12*	0.21**	0.22***	0.20***
	(1.57)	(1.66)	(1.68)	(2.51)	(3.15)	(2.81)
Legal Proceedings	0.36**	0.37***	0.33***	0.28	0.30**	0.25*
	(2.24)	(3.09)	(2.70)	(1.57)	(2.36)	(1.93)
Quant. and Qual. Disclosures about Market Risk	0.69***	0.68***	0.68***	0.20**	0.21***	0.19***
	(2.75)	(2.69)	(2.65)	(2.37)	(2.96)	(2.60)
Risk Factors	1.14	1.18	1.18	1.43**	1.44**	1.88***
	(1.61)	(1.63)	(1.64)	(2.13)	(2.45)	(2.76)
Other Information	0.20	0.27	0.36*	0.31*	0.37**	0.40**
	(1.08)	(1.47)	(1.92)	(1.78)	(2.19)	(2.30)

(*Continued*)

Lazy Prices

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(b) Table VII: section returns

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Table VII—*Continued*

Panel A: Equally Weighted						
	Sim_MinEdit			*Sim_Simple*		
	Excess Return	Three-Factor Alpha	Five-Factor Alpha	Excess Return	Three-Factor Alpha	Five-Factor Alpha
Management's Discussion and Analysis	0.18*	0.22***	0.19***	0.19***	0.19**	0.17**
	(1.95)	(3.16)	(2.67)	(2.67)	(2.54)	(2.32)
Legal Proceedings	0.22	0.25**	0.22*	0.13	0.16	0.12
	(1.27)	(2.30)	(1.93)	(0.82)	(1.41)	(1.10)
Quant. and Qual. Disclosures about Market Risk	0.16	0.23*	0.22*	0.13	0.11	0.07
	(1.18)	(1.74)	(1.67)	(0.16)	(0.13)	(0.08)
Risk Factors	1.02	1.85***	1.38**	1.25*	1.54**	1.77**
	(1.19)	(2.77)	(2.17)	(1.93)	(2.19)	(2.12)
Other Information	0.09	0.14	0.16	0.22	0.26**	0.22*
	(0.58)	(0.97)	(1.05)	(1.27)	(2.31)	(1.95)

(Continued)

(a) Table VII continued

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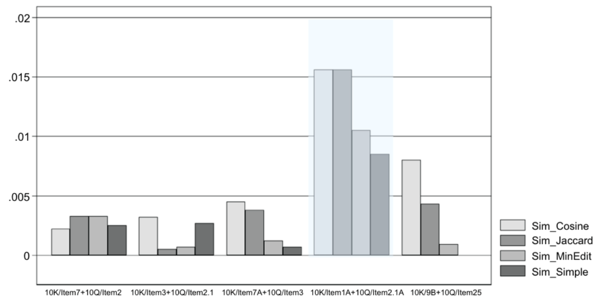

Figure 9. Five-factor alphas for portfolio sort, by important common sections for 10-Ks and 10-Qs. This figure shows the five-factor alphas (market, size, value, momentum, and liquidity) of the top (highest similarity) minus bottom (lowest similarity) quintile portfolio (Q5 – Q1) for the common sections of a firm's 10-K and 10-Q financial reports: Management's Discussion and Analysis, Legal Proceedings, Quantitative and Qualitative Disclosures about Market Risk, Risk Factors, and Other Information:
10-K/Item7 + 10-Q/Item2: Management's Discussion and Analysis of Financial Condition and Results of Operations
10-K/Item3 + 10-Q/Item2.1: Legal Proceedings
10-K/Item7A + 10-Q/Item3: Quantitative and Qualitative Disclosure about Market Risk
10-K/Item1A + 10-Q/Item2.1A: Risk Factors

(b) Figure 9: section alphas

6.6 Tables VIII–XI: Attention, comparative statements, real effects, and robustness

Read: Return predictability weakens when more investors compare filings and when firms use explicit comparative statements. Similarity also predicts future operating outcomes and remains robust to known return predictors.

Economic message. The evidence points to investor inattention and delayed incorporation of real information.

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Table VIII
Interacting with Investor Attention

This table reports results of Fama-MacBeth cross-sectional regressions of individual firm-level stock returns on our similarity measures and interactions of the similarity measures with *IPAccessMultipleYear*. *Return*, the dependent variable, is multiplied by 100. *IPAccessMultipleYear* is a proxy for firms with investors who do check the changes in 10-Ks/10-Qs and is given as the number of unique IP addresses that access both the current 10-K/10-Q and previous year's 10-K/10-Q for the same firm normalized by the total number of unique IP addresses that access the current 10-K/10-Q. We download EDGAR traffic log file from the SEC and remove robot requests as in Loughran and McDonald (2017). t-Statistics are reported below the estimates. Statistical significance at the 1%, 5%, and 10% levels is indicated by ***, **, and *, respectively.

	Dependent Variable: Return							
	Sim_Cosine		*Sim_Jaccard*		*Sim_MinEdit*		*Sim_Simple*	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Similarity	0.44**	0.42**	0.78***	0.84***	0.65***	0.73***	0.06**	0.06**
	(2.56)	(2.37)	(2.90)	(3.08)	(2.70)	(2.94)	(2.13)	(2.30)
IPAccessMultipleYear × *Similarity*		−0.27		−0.84**		−0.79*		−0.10**
		(−0.65)		(−2.08)		(−1.73)		(−2.05)
IPAccessMultipleYear		0.11		0.15		0.11		0.08**
		(0.31)		(0.86)		(0.50)		(2.05)
Cons	0.52	0.54	0.59	0.57	0.65	0.63	−0.04	−0.04*
	(1.16)	(1.20)	(1.36)	(1.31)	(1.50)	(1.44)	(−1.53)	(−1.70)
R^2	0.0006	0.0014	0.0016	0.0024	0.0017	0.0025	0.0019	0.0027
N	547,918	547,918	547,918	547,918	547,918	547,918	548,912	548,912

(a) Table VIII: attention interactions

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Table IX
Explicitly Comparative Statements

This table reports results of calendar-time portfolio value-weighted five-factor alphas (market, size, value, momentum, and liquidity) for samples of firms that explicitly make comparative statements in their annual and quarterly filings versus firms that do not. Portfolio returns are multiplied by 100. Firms that make explicit comparisons are taken to be those whose financial reports include phrases listed in Panel B. More specifically, we search for instances in which a word/phrase from Group A is within 10 words of a word/phrase from Group B. For each subsample, we compute quintiles based on the prior year's distribution of similarity scores across all stocks. Stocks then enter quintile portfolios in the month after the public release of one of their 10-K or 10-Q reports. Firms are held in quintile portfolios for three months. t-Statistics are reported below the estimates. Statistical significance at the 1%, 5%, and 10% levels is indicated by ***, **, and *, respectively.

Panel A: Alphas across Firms Making (Not Making) Explicit Comparison Statements in Year-over-Year Documents						
Explicit Comparative Statements	Five-Factor Alpha, Jaccard Similarity					
	Q1	Q2	Q3	Q4	Q5	Q 5 – Q1
Yes	0.22	−0.24	−0.06	0.22	0.31	0.09
	(1.04)	(−0.84)	(−0.29)	(1.11)	(1.54)	(0.34)
	Q1	Q2	Q3	Q4	Q5	Q 5 – Q1
No	−0.36***	−0.07	−0.07	0.06	0.17	0.53***
	(−3.39)	(−0.57)	(−0.59)	(0.55)	(1.57)	(3.51)

Panel B: Example Phrases Captured in 10-Ks and 10-Qs		
Group A	+	Group B
Sales		Last year
EBITDA		Prior year

(b) Table IX: comparative statements

Table X

Real Effects

This table reports results of regressions of operating income, net income, and sales on a firm's lagged similarity measures. *Oibdpq / L1atq* is operating income before depreciation (Oidbpq) divided by lagged total assets (L1atq). *Niq / L1atq* is net income (Niq) divided by lagged total assets (L1atq). *Saleq / L1atq* is sales (Saleq) divided by lagged total assets (L1atq). Dependent variables are multiplied by 100. All variables in the table are winsorized at the 1% level. All regressions include month, industry, and firm fixed effects. Standard errors are adjusted for clustering at the monthly level. *t*-Statistics calculated using robust clustered standard errors are reported in parentheses. Statistical significance at the 1%, 5%, and 10% levels is indicated by ***, **, and *, respectively.

	Oibdpq / L1atq				*Niq / L1atq*				*Saleq / L1atq*			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
Sim_Cosine	0.50*				0.48				0.01*			
	(1.96)				(1.44)				(1.95)			
Sim_Jaccard		0.68***				0.89***				0.01***		
		(10.68)				(10.48)				(7.83)		
Sim_MinEdit			0.65***				0.75***				0.02***	
			(12.48)				(10.89)				(14.48)	
Sim_Simple				0.51***				0.71***				0.01***
				(7.80)				(8.41)				(6.85)
Cons	−0.01***	−0.40***	−0.01***	−0.02***	−0.04***	−0.04***	−0.04***	−0.04***	0.21***	0.22***	0.22***	0.19***
	(−4.71)	(−3.05)	(−8.59)	(−6.33)	(−11.17)	(−24.07)	(−23.57)	(−12.76)	(27.33)	(51.47)	(53.73)	(27.67)
Month FEs	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Industry FEs	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Firm FEs	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
R^2	0.0585	0.0116	0.2858	0.0558	0.0581	0.0549	0.2859	0.0588	0.0596	0.0563	0.287	0.2864
N	284,151	284,151	284,151	325,717	295,031	295,031	295,031	338,477	295,031	295,031	295,031	338,476

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(a) Table X: real effects

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Table XI

Robustness—Fama-MacBeth with More Controls

This table reports results of Fama-MacBeth cross-sectional regressions of individual firm-level monthly stock returns on our four similarity measures and a number of known return predictors. *Return*, the dependent variable, is multiplied by 100. *Size* is log of market value of equity, *log*(*BM*) is log book value of equity over market value of equity, *Ret*(−1,0) is previous month's return. *Ret*(−3,–1), *Ret*(−6,–1), *Ret*(−9,–1), *and Ret*(−12,–1) are the cumulative return from month −3 to month −1, month −6 to month −1, month −9 to month −1, and month −12 to month −1, respectively. *Invest* is capx/ppent. *GrossProfit* is (revt-cogs)/at. *FreeCashFlow* is (ni + dp − wcapch − capx)/at. *Accrual* is (Δact − chech − Δlct + Δdct + Δtxp − dp) scaled by average assets (at/2 + lag(at)/2). SUE is Compustat-based standardized unexpected earnings, computed as in Livnat and Mendenhall (2006). The Compustat earnings surprise is based on the assumption that EPS follows a seasonal random walk, where the best expectation of the EPS in quarter *t* is the firm's reported EPS in the same quarter of the previous fiscal year. *t*-Statistics are reported below the estimates. Statistical significance at the 1%, 5%, and 10% levels is indicated by ***, **, and *, respectively

	Ret			
	(1)	(2)	(3)	(4)
Sim_Cosine	0.38***			
	(3.19)			
Sim_Jaccard		0.55***		
		(4.20)		
Sim_MinEdit			0.35***	
			(3.06)	
Sim_Simple				3.18**
				(2.39)
Ret(−1,0)	−2.98***	−3.00***	−3.01***	−2.95***
	(−5.77)	(−5.81)	(−5.83)	(−5.58)
Ret(−3,–1)	0.00	−0.01	0.00	−0.05
	(−0.01)	(−0.01)	(−0.01)	(−0.11)
Ret(−6,–1)	(0.06)	(0.05)	−0.05	0.01
	(0.17)	(0.16)	(0.15)	(0.03)
Ret(−12,–1)	0.57**	0.57**	0.56**	0.59**
	(2.41)	(2.40)	(2.40)	−2.48
Size	0.00	0.01	0.01	−0.01
	(0.03)	(0.14)	(0.16)	(−0.19)
log(*BM*)	0.12**	0.13**	0.13**	0.12*
	(2.02)	(2.06)	(2.07)	(1.90)
Invest	−0.26	−0.24	−0.25	−0.23
	(−0.81)	(−0.75)	(−0.77)	(−0.69)
GrossProfit	0.33*	0.32*	0.32*	0.3
	(1.88)	(1.84)	(1.82)	(1.63)
Accrual	−0.98***	−0.98***	−0.98***	−1.07***
	(−4.16)	(−4.18)	(−4.17)	(−4.62)
FreeCashFlow	0.84**	0.80**	0.81**	0.86**
	(2.31)	(2.22)	(2.25)	(2.33)
SUE	0.11***	0.11***	0.11***	0.11***
	(5.53)	(5.55)	(5.57)	(4.88)

(b) Table XI: robustness

7 Short summary

- Year-over-year filing changes predict future returns and fundamentals.
- There is little filing-date reaction, so the evidence points to inattention rather than ordinary underreaction.
- EDGAR download logs and comparative language tests support the mechanism.
- Risk Factors and MD&A are especially informative sections.
- The findings show that public information can remain economically unprocessed.

8 Conclusion

8.1 Core conclusion

The paper shows that corporate filings remain highly informative, but investors often miss changes at the time of release. The information is incorporated into prices only gradually.

8.2 Incremental contribution

The contribution is to transform document change into a priced signal and to connect that signal to direct measures of investor comparison behavior.

8.3 Mechanism

The absence of announcement returns, the EDGAR attention interaction, and the comparative-statement evidence all point to processing inattention.

8.4 Implications

For investors, systematic comparison of current and prior filings can uncover delayed information. For regulators and firms, explicit change disclosure can improve price discovery.

8.5 Takeaway

The paper's lasting insight is that information abundance can create lazy prices: disclosure is public, but comparison and interpretation remain scarce.